

# **MY FINAL PROJECT**

**By: Connor Beachler**

# MY IDEA

- Create a program that allows the teacher to compete with his students in a game and then sends that student to a fate.

# MY GAME CLASS

- Rock, paper, scissors
- Guess the number
- Tic Tac Toe
- Teacher V Student

```
94  private boolean ticTacToe() {
95      // Creates a board for the game
96      char[] board = {
97          '1','2','3',
98          '4','5','6',
99          '7','8','9'
100     };
101
102     // Sets up a player as X
103     char currentPlayer = 'X';
104     boolean teacherTurn = true;
105
106     // Goes through the board until complete
107     while (true) {
108         // Shows the board
109         printBoard(board);
110
111         // Lets the teacher go first then the student
112         if (teacherTurn) {
113             System.out.print("Teacher choose spot: ");
114         }
115         else {
116             System.out.print("Student choose spot: ");
117         }
118
119         int move = Integer.parseInt(keyboard.nextLine()) - 1;
120
121         // Makes sure that all the moves are valid
122         if (move < 0 || move > 8 || board[move] == 'X' || board[move] == 'O') {
123             System.out.println("Invalid move.");
124             continue;
125         }
126
127         board[move] = currentPlayer;
128
129         // Checks to see if there is a winner yet and if so prints it out
130         if (checkWinner(board, currentPlayer)) {
131             printBoard(board);
```

# MY STUDENT MANAGER CLASS

- Made sure that either someone gets a student randomly
- Or the student they chose is in the list

```
class StudentManager {  
    // Creates a array list of the students  
    private ArrayList<String> students = new ArrayList<>();  
  
    // Creates a student manager function  
    public StudentManager(String filename) {  
        // Tries to get the file and catch it if it's not found  
        try {  
            File file = new File(filename);  
            Scanner keyboard = new Scanner(file);  
  
            while (keyboard.hasNextLine()) {  
                students.add(keyboard.nextLine());  
            }  
  
            keyboard.close();  
        } catch (FileNotFoundException e) {  
            System.out.println("Error: File not found.");  
        }  
    }  
  
    // Creates a function that gets a random student  
    public String getRandomStudent() {  
        Random rand = new Random();  
  
        // Checks to see if there are any students found  
        if (students.isEmpty()) {  
            return "No Students Found";  
        }  
  
        // Tries to get a random student  
        return students.get(rand.nextInt(students.size()));  
    }  
  
    // Function that checks if a student is in the list or not  
    public boolean containsStudent(String name) {  
        return students.contains(name);  
    }  
}
```

# MY EXILE CLASS

- Randomly decides what happens to a student if they lose the game

```
import java.util.Random;

// Creates the exile class
class Exile {
    // Creates a string of punishments
    private String[] punishments = {
        "Life in detention",
        "Expelled!",
        "Forced to use Rust only",
        "Banished to Siberia",
    };

    // Creates a string of rewards
    private String[] paradise = {
        "A perfect 100!",
        "A million dollars in your bank account!",
        "Full paid vacation to your location of choice!",
        "You get to graduate right now!",
    };

    private Random rand = new Random();

    // Creates a function to exile a student
    public void exileStudent(String name) {

        // Makes an exception for Connor Beachler only
        if (name.equalsIgnoreCase("Connor Beachler")) {
            String reward = paradise[rand.nextInt(paradise.length)];
            System.out.println("\n" + name + " " + reward + "!");
            return;
        }

        // Makes sure to show the punishment for each student and randomly select one
        String fate = punishments[rand.nextInt(punishments.length)];
        System.out.println("\n" + name + " " + fate + "!");
    }
}
```

# MAIN

```
public class Main {
    public static void main(String[] args) {
        // Creates the scanner variable and the variable for the game, student manager, and the exile classes
        Scanner keyboard = new Scanner(System.in);
        StudentManager manager = new StudentManager("students.txt");
        Game game = new Game();
        Exile exile = new Exile();
        String student = "";

        // This is a welcome message for Adam Spanier
        System.out.print("Hello Adam Spanier!\nHave your students been bad? Well it's time to finally get your sweet " +
            "revenge!\nWe will allow you to either pick the student you want punished or simply make an example out" +
            " of one of them randomly! Please enter which decision you would rather do!");

        String a = "";
        // Creates a while statement to keep the program running until the user wants out
        while (!a.equalsIgnoreCase("q")) {
            // Asks the user for their input
            System.out.println("\n\t1: Randomly make an example out of a student\n\t2: Show that one irritating student" +
                " that YOU are in charge and make him or her PAY for their transactions\n\t0r Enter 'q' to quit");
            a = keyboard.nextLine();
            // Two random responses to two possible outside the box answers to the prompt
            if (a.equals("67")) {
                System.out.println("Okay this joke is like a year old get over it");
            }
            if (a.equals("3")) {
                System.out.println("To access the third option DLC please enter your credit card number below: \n");
                String b = keyboard.nextLine();
                System.out.println("Jk, if you actually put your information however Adam it's time to retire");
            }
            // Tells the user to actually answer the question
            if (!a.equals("1") && !a.equals("2") && !a.equalsIgnoreCase("q")) {
                System.out.println("Come on Adam!! There are two options (1 or 2) not whatever you selected!");
            }
            // Asks the user for a student input and if it doesn't get it. It gets a random student instead
            if (a.equals("2")) {
                System.out.println("\nWho's it going to be????");
                student = keyboard.nextLine();

                if (!manager.containsStudent(student)) {
                    System.out.println("Student not found. Using random instead");
                    student = manager.getRandomStudent();
                }
                System.out.println("\nSelected student: " + student);

                // Makes the student play the game with the teacher
                boolean teacherWon = game.playGame();
            }
        }
    }
}
```

- Puts everything together with an introduction
- Allows for a random or a selected student
- Allows for the teacher to choose the game
- Allows for student v teacher gameplay
- Banishment or victory